AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 2 · STUDENT LESSON

Input, Process, Output

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Follow the information. Notice what is lost.

Start with a library kiosk, trace the pencil-box summary, practise with visitor records, then explain a new case.

Keep one habit: Name the exact input → trace the transformation → state what the output cannot tell you.

MAIN LESSON · SECTION 1

What happened to the information?

> Trace what enters, what changes and what comes out. Then ask which details were kept.

Start with a puzzle

A screen shows three book titles. Has the computer spoken, reserved a book, or read a whole story? This week, follow the information one step at a time before crediting a system with extra actions.

Reconnect with Week 1

In last week's bus case, the forecast number and the display colour came from different decisions. Which decision produced each **output**?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can **trace** information through a system without confusing information with a physical action.
- I can explain which details a **representation** preserves or discards.
- I can locate the earliest stage at which an incorrect result could have arisen.

<details>
<summary>Earlier wording, alternative practice and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

A system transforms information; it does not simply 'know.'
</details>



LEARNING TARGET A library kiosk receives 'mystery', filters a catalogue by genre and displays three titles. No speaker is attached. Draw its information path. A visitor says it read three books aloud.

Which stage contradicts that claim?

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can **trace** an information path and name what a summary loses.

MAIN LESSON · SECTION 2

A title on a screen is not a spoken book

> Read the supplied facts before deciding what the kiosk has done.

Meet Sam at the library kiosk

A library kiosk asks Sam for a genre. Sam selects mystery. The kiosk compares that preference with catalogue tags and displays three book titles. Sam says the kiosk has read three books aloud.

That conclusion adds an action that the description never supplied: displaying a title is different from producing spoken audio.

Follow the information

An information path names what enters each stage, what happens there and what leaves it. Treat **input** and **output** as roles relative to a stage.

A list of titles is the **output** of the search, but it could become the **input** to a separate text-to-speech stage. A component can therefore receive the **output** of another component without doing the same job.

Connect the mechanism to Week 1

The kiosk's genre-matching operation can be implemented with a **written rule**. Nothing in this description establishes that a model was trained.

This connects to Week 1: tracing information tells us what a **process** needs to do, while **mechanism evidence** tells us how it does it.

Explain the difference

- Name three visible details without explaining them yet.

- Make a first **prediction** and state the clue that supports it.

▼ Earlier wording, alternative practice and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A music helper receives listening choices, compares them with a ranking **process**, and suggests a song. The suggestion appears only after information has moved through the system.

What do you notice?

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

One stage’s output can become another stage’s input

> Name the actual content at each arrow, not just the word “text”.

Read the three possible stages

Read each row as its own possible stage. The first row is the supplied kiosk. The speech and borrowing rows describe extra stages that would need to exist and run; they are not evidence that this kiosk performed them.

The kiosk and two possible additional stages

INPUT	PROCESS	OUTPUT
Genre request mystery and catalogue tags	Keep matching catalogue records	Three matching titles
Three titles from the first stage	A separate speech system if available	Spoken titles rather than whole books
A selected title and an available copy	A person completes the borrowing process	A recorded loan and a borrowed copy

The first row is the kiosk as described. The later rows require additional components and actions. Arrows show information dependencies; they do not establish that every stage has actually run.

A title is smaller than a whole book

Notice the difference between a title, a description and an entire book. All three can be text, but they represent different amounts of information.

Even a speech stage receiving three titles can speak only those supplied titles; it has not thereby read the full books. Always identify the content of an **input**, not just its data type.

A suggestion is different from a loan

A recommendation is also different from a loan. Selecting a title does not reserve a copy unless an authorised reservation step exists.

Keeping these distinctions clear prevents us from crediting a system with actions it did not take.

Connect to another system

UNDER THE HOOD **Input**, **process**, and **output** are functional roles. The **process** may contain human choices, fixed rules, calculations, retrieval, or a **learned model**.

Naming the three stages prevents magical explanations of computer behavior.

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternative practice and picture transcripts

VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



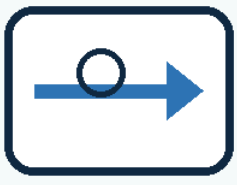
INPUT

songs played, liked, or skipped



PROCESS

compare clues and rank possible songs



OUTPUT

a suggested song or a not-sure result

INPUT songs played, liked, or skipped

PROCESS compare clues and rank possible songs

OUTPUT a suggested song or a not-sure result

CORE IDEA A system transforms information; it does not simply 'know.'

Draw a fresh example with all three stages:

A closer explanation

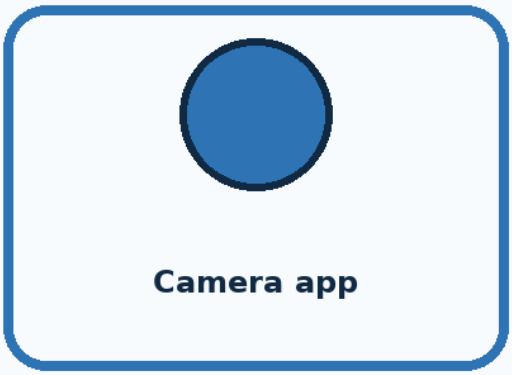
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare the roles, not just the pictures

> Input and output depend on which stage you are describing.

Four clues, paired with their explanations



The case to investigate

CASE 1 — Camera app Clue: photo enters before any result. Meaning: **input**.



The case to investigate

CASE 2 — Music ranking Clue: choices are compared and scored. Meaning: **process**.



The case to investigate

CASE 3 — Song suggestion Clue: a result is shown to the user. Meaning: **output**.



The case to investigate

CASE 4 — No listening history Clue: important **input** is missing. Meaning: not-sure case.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep a limit in view

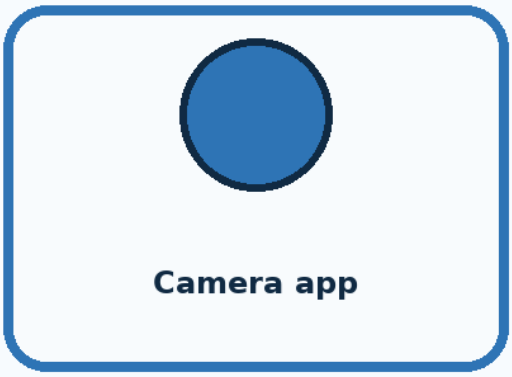
BOUNDARY The **output** is not proof that the system understood the **input** as a person would.

▼ Earlier wording, alternative practice and picture transcripts

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Camera app Clue: photo enters before any result. Meaning: **input**.

CASE 2 — Music ranking Clue: choices are compared and scored. Meaning: **process**.

CASE 3 — Song suggestion Clue: a result is shown to the user. Meaning: **output**.

CASE 4 — No listening history Clue: important **input** is missing. Meaning: not-sure case.

Change one thing in the pictured case

Change: Store red count alongside total.

Prediction: The records become (6,2) and (6,5).

Reason: Blue count can now be recovered by total minus red, assuming these are the only colours.

MAIN LESSON · SECTION 6

Two different boxes. The same stored total.

> A correct total can still leave out the detail needed for your next question.

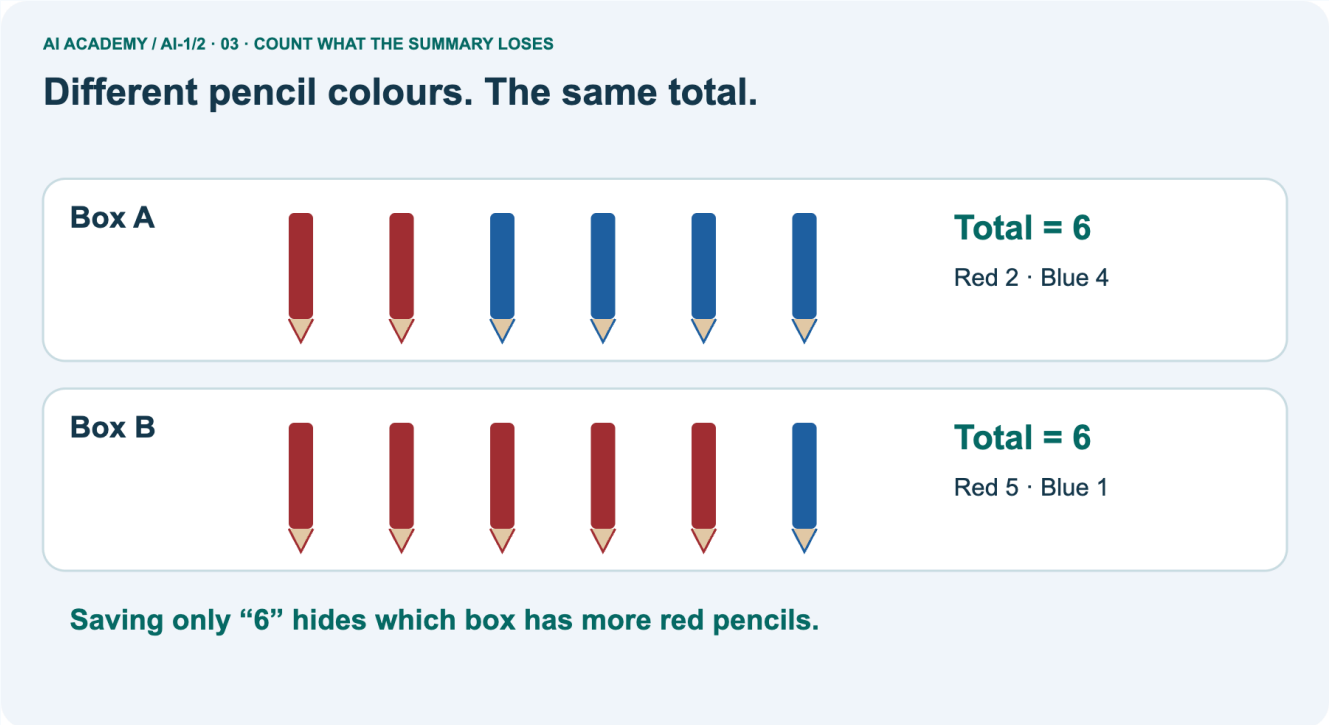
The case and its given facts

A fictional classroom cupboard records coloured pencils, then reports just a total.

Box A has 2 red and 4 blue pencils. Box B has 5 red and 1 blue.

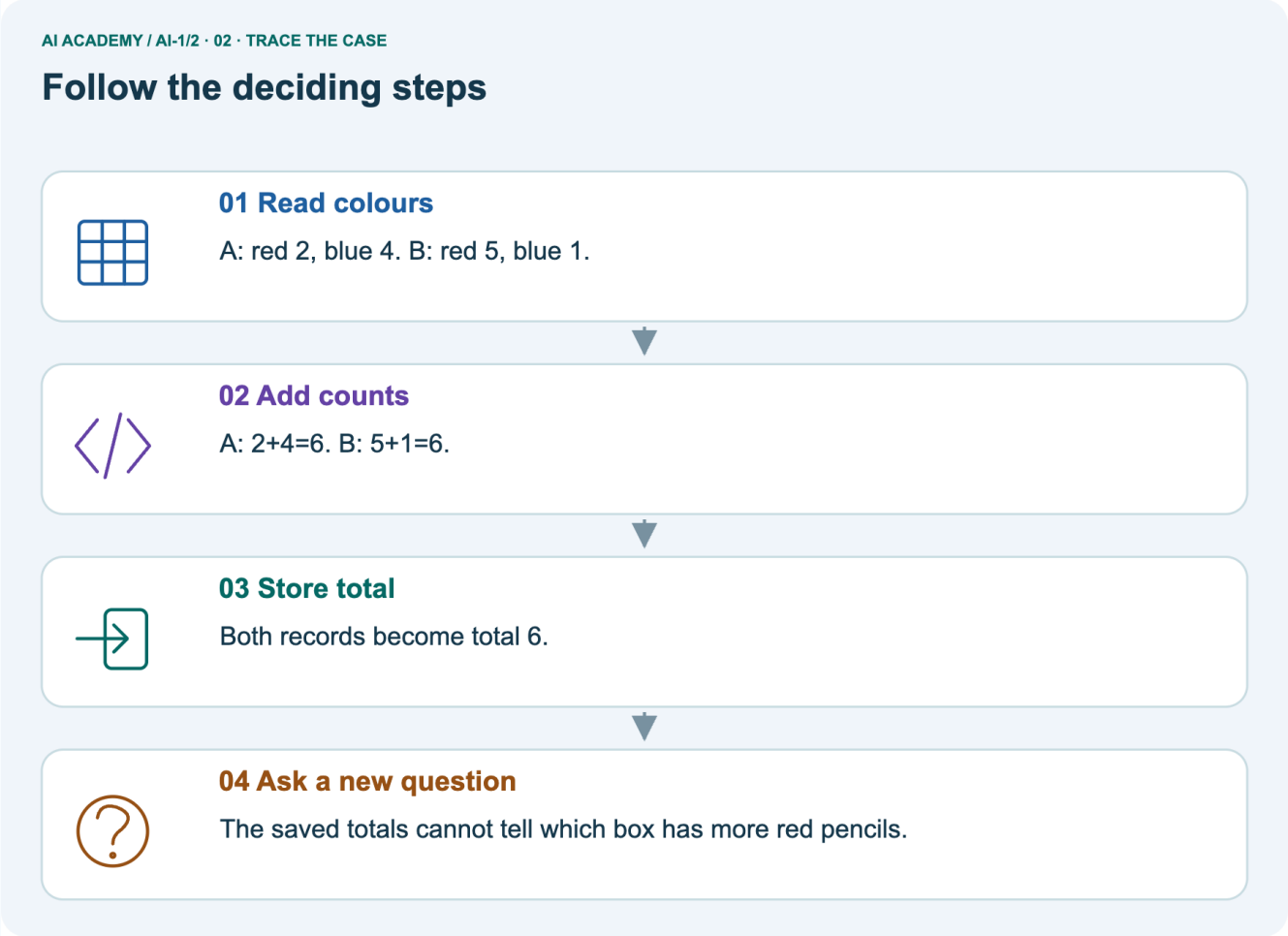
The **process** adds colour counts and stores only the total, not the two counts.

Compare the original boxes



Follow the labelled picture with the matching explanation. Open the vector diagram

Follow what the device keeps



Follow the labelled picture with the matching explanation. Open the vector diagram

Why the colour counts cannot be recovered

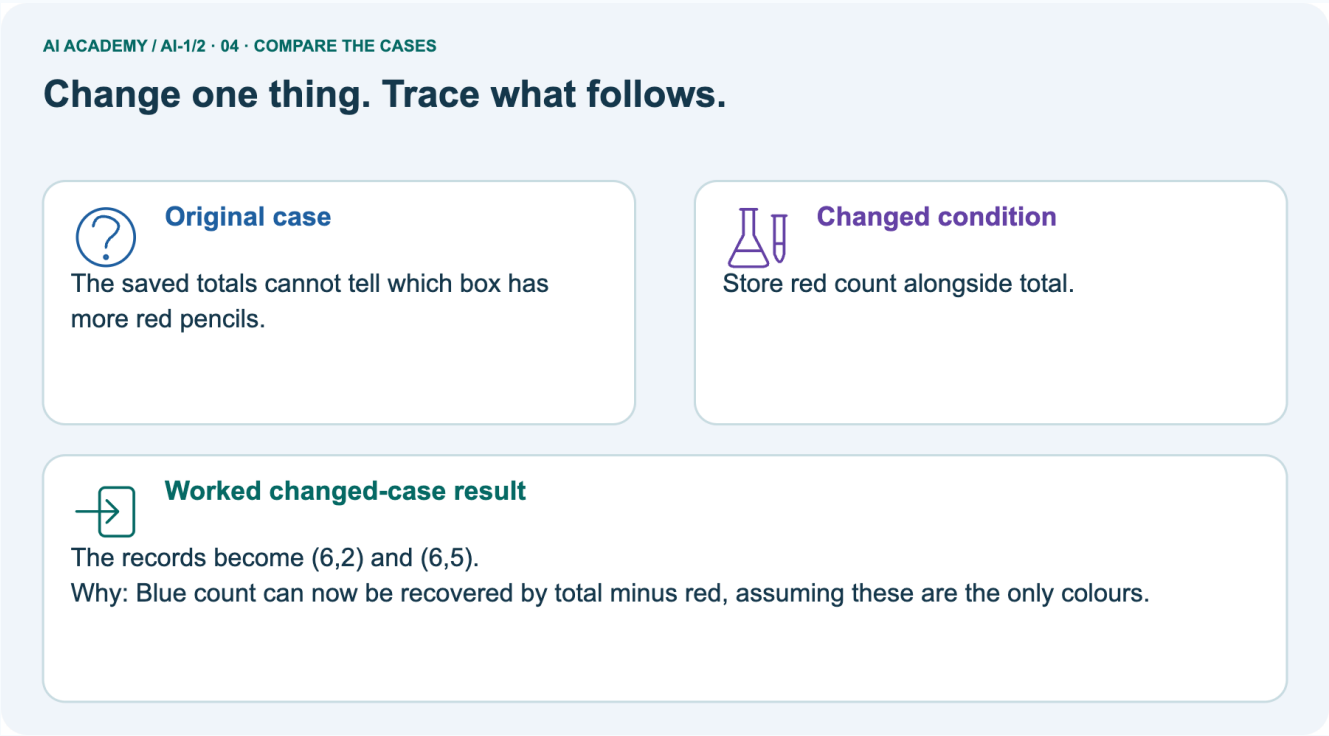
Adding preserves the number of pencils but discards how that number was divided between colours. Both distinct **input**s lead to the same stored **output**.

Working backwards from 6 therefore has several possible answers; it is not an instruction to divide equally.

Predict a richer record

- Before opening the changed-case picture: suppose the device stores total AND red count. What would each box's record be? Could you now find its blue count? Try it using the two-colour assumption.

▼ Check the changed record



Follow the labelled picture with the matching explanation. Open the vector diagram

Respect the assumption

The two-colour assumption matters. Extra colours would prevent inferring blue from total and red alone.

Use the richer record

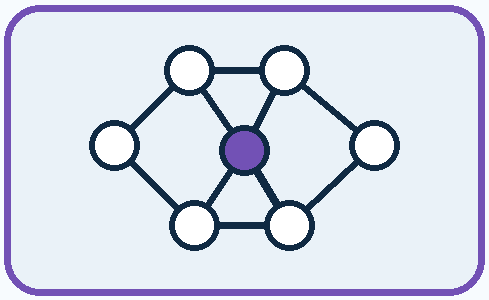
- Read the labelled picture: Use the richer record to find blue pencils in Box B. Explain why total alone was insufficient.

▼ Earlier wording, alternative practice and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A music helper receives listening choices, compares them with a ranking **process**, and suggests a song. The suggestion appears only after information has moved through the system.

STEP 1 — OBSERVE songs played, liked, or skipped

STEP 2 — **TRACE** compare clues and rank possible songs

STEP 3 — CONCLUDE a suggested song or a not-sure result

MODEL REASONING Start with what enters, name the actual transformation rather than saying the system thinks, and identify the result. If the **process** is unknown, state what **evidence** is missing instead of inventing a **mechanism**.

DO NOT OVERCLAIM The **output** is not proof that the system understood the **input** as a person would.

What new **evidence** could change this answer?

A closer explanation

Consider two catalogue records. Book A has 90 pages and is about a child solving a seaside mystery. Book B has 90 pages and is about a lost spaceship. If we keep only page count, both become the number 90.

The number is useful for a length limit, but it cannot tell us which story is a mystery. Information that was never retained cannot be recovered just by calculating harder.

Representation kept	Question it can help answer	Question it cannot settle alone
Page count 90	Is the book under 100 pages?	Is its story a mystery?
Genre tag mystery	Does it match a mystery request?	Will this particular reader enjoy it?
Short summary	What does the description say happens?	What exact sentence occurs on page 42?

The phrase cannot settle alone matters. A genre tag might be inaccurate, and a reader might enjoy an unfamiliar genre. A **representation** is useful only in relation to a question and its limits.

More detail is not automatically better: unnecessary personal information can add risk without helping the task.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Input, Process, Output

The facts we will use

AI ACADEMY / AI-1/2

Different colour counts, same total

	Red	Blue	Total
Box A	2	4	6
Box B	5	1	6

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/2

Input, Process, Output

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read colours

A: red 2, blue 4. B: red 5, blue 1.



2. Add counts

A: 2+4=6. B: 5+1=6.



3. Store total

Both records become total 6.



4. Ask a new question

The saved totals cannot tell which box has more red pencils.

CHANGE ONE THING

Store red count alongside total.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read colours	A: red 2, blue 4. B: red 5, blue 1.
2. Add counts	A: 2+4=6. B: 5+1=6.
3. Store total	Both records become total 6.
4. Ask a new question	The saved totals cannot tell which box has more red pencils.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: Store red count alongside total. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Find the first wrong stage

> A correct calculation can still use an incorrectly recorded input.

Investigate a wrong catalogue result

The kiosk is asked for mystery books under 100 pages but displays a 140-page mystery. Before changing everything, inspect the intermediate records. Was the maximum entered as 100? Did the catalogue say 140?

Did filtering preserve the wrong record? Did the display show a title different from the filtered result? The first mismatch between the intended and actual path helps locate the problem.

Check intermediate values

A correct-looking final screen can hide an earlier error. A calculator can correctly multiply a number that was transcribed incorrectly, and a warning rule can correctly use an incorrect forecast.

Write down intermediate values so you can test each connection.

Repair two tempting claims

- CLAIM A The computer already knows the answer before it receives **input**.
- CLAIM B The screen is the **process** because that is where the result appears.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

Explain why a system that displays a book title has not necessarily spoken, reserved or read the book

▼ **Earlier wording, alternative practice and picture transcripts**

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Camera app	photo enters before any result	input
Music ranking	choices are compared and scored	process
Song suggestion	a result is shown to the user	output
No listening history	important input is missing	not-sure case

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Practise: what can one number tell us?

> Construct two possible originals that share a summary but answer a question differently.

Model the method with temperatures

A greenhouse records morning temperatures of 18, 22 and 26 degrees C, then retains only the largest value, 26. Can the stored value prove that the temperature rose steadily? No.

The sequence 22, 18, 26 has the same largest value but includes a decrease. These two sequences form a counterexample to the claim that the largest value determines the order.

To solve this, first state exactly what was kept: one maximum. Then ask what the claim requires: the order of several readings.

Finally construct two possible originals that share the retained value but give different answers to the claim. If both originals fit the record, the record alone cannot decide between them.

Now try the visitor record

1. Your W2 case: only total visitors=3 was kept. Write two possible arrival schedules with that same total. Use their different busiest-hour patterns to explain what the total cannot tell you.

2. Use workbook W2 for the connected example "**Representation**s and information paths". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

A record keeps only a visitor count of 3. Give two different arrival schedules consistent with it and name one fact the count cannot reveal

Explain, check and switch roles

Explain your **trace** to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge “every output is the final action”

- Use the worked case for **Representation**s and information paths. Proposed explanation: “Every **output** is the final action”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check the pencil explanation

Use the richer record to find blue pencils in Box B. Explain why total alone was insufficient.

> **My explanation using a pictured step and the given facts**

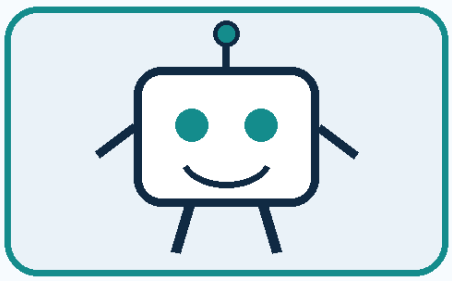
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternative practice and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A system transforms information; it does not simply 'know.'

Use a fresh example to explain **input**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: The screen is the **process** because that is where the result appears.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Finish the visitor example

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate a runnable music helper

> Use a supplied rule so your prediction, trace and explanation can be checked.

Your fictional system

Use this fictional paper music helper. Input: one mood word. Rule: “calm” returns Quiet River; “lively” returns Jumping Feet; a missing or unrecognised mood returns “Please choose calm or lively.” Its output is a displayed title or request, not audio. These are toy titles, and this rule is not a trained model.

Predict, trace and swap

1. In pairs, predict the result for calm. Then let your partner apply the written rule and record the paper-trace result. Swap roles and try a missing mood. Keep the predicted result separate from what your partner actually produced; a paper trace is not a run of an AI model.
2. Keep a normal case and an edge case clearly separated.
3. Record what happened—even when it does not match your **prediction**.
4. Explain the result, one limitation, and the next useful test.

Record normal and missing-input cases

Case	Prediction	Observed result	What it means
Normal			
Edge			

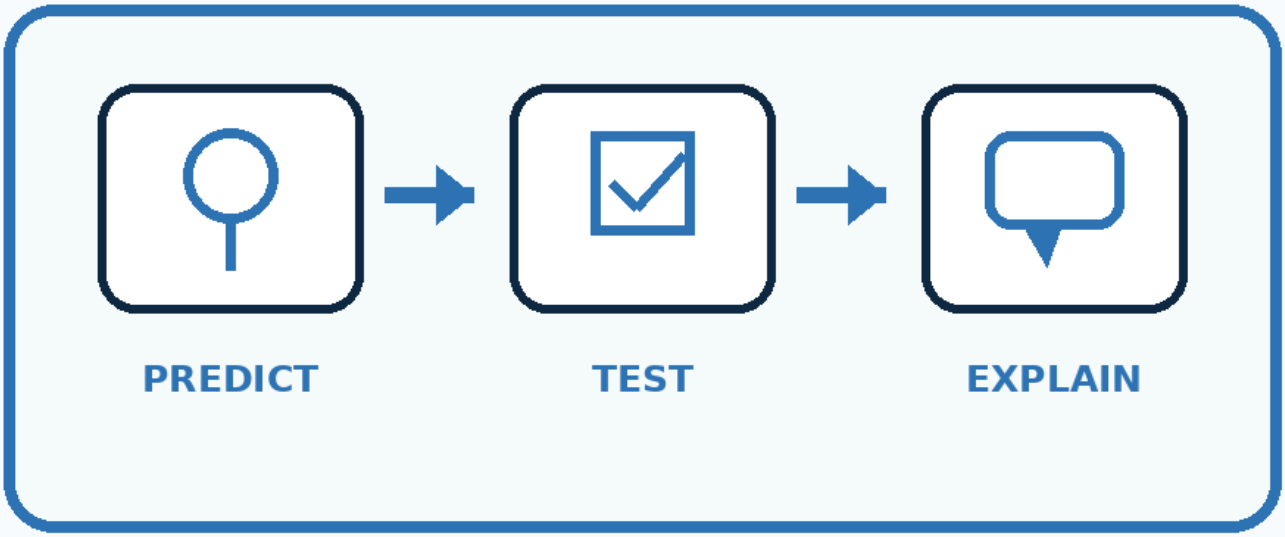
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only **trace**d the procedure.

▼ Earlier wording, alternative practice and picture transcripts

EXPLORER LAB

Human Music Recommender

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: **Trace** a music recommender and draw all three stages.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: does a total reveal every score?

> Keep one first attempt before hints or teacher feedback.

The new case

A classroom device receives four scores and stores only their total, 20. A student claims that every score must be 5. Give two different lists of four nonnegative whole-number scores with total 20.

Explain precisely what the stored total establishes and what it does not.

My independent explanation, trace and limits

Keep the attempt fair

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after your first attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternative practice and picture transcripts**

Independent case: Investigate a new loss of information

Week 2 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A classroom fan receives a temperature reading, compares it with 25 degrees C, and turns on. Name the **input**, **process** and **output**. What should happen if the reading is missing?

My answer, with a **trace**, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Draw the Classroom Sorter's first **input**-**process**-**output** map and show where a person may review the result.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A library kiosk receives 'mystery', filters a catalogue by genre and displays three titles. No speaker is attached. Draw its information path. A visitor says it read three books aloud. Which stage contradicts that claim?

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Build your sorter’s information map

> Keep the information your decision needs and name the details you leave out.

This week’s design decision

THIS WEEK'S CHECKPOINT Draw the Classroom Sorter's first **input**-**process**-**output** map and show where a person may review the result.

Choose sorter **input**s that retain the clues needed for its decision; draw the transformation and identify discarded details.

Plan the stages and review point

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?

- When should it say not sure or ask a person?

Avoid a larger claim

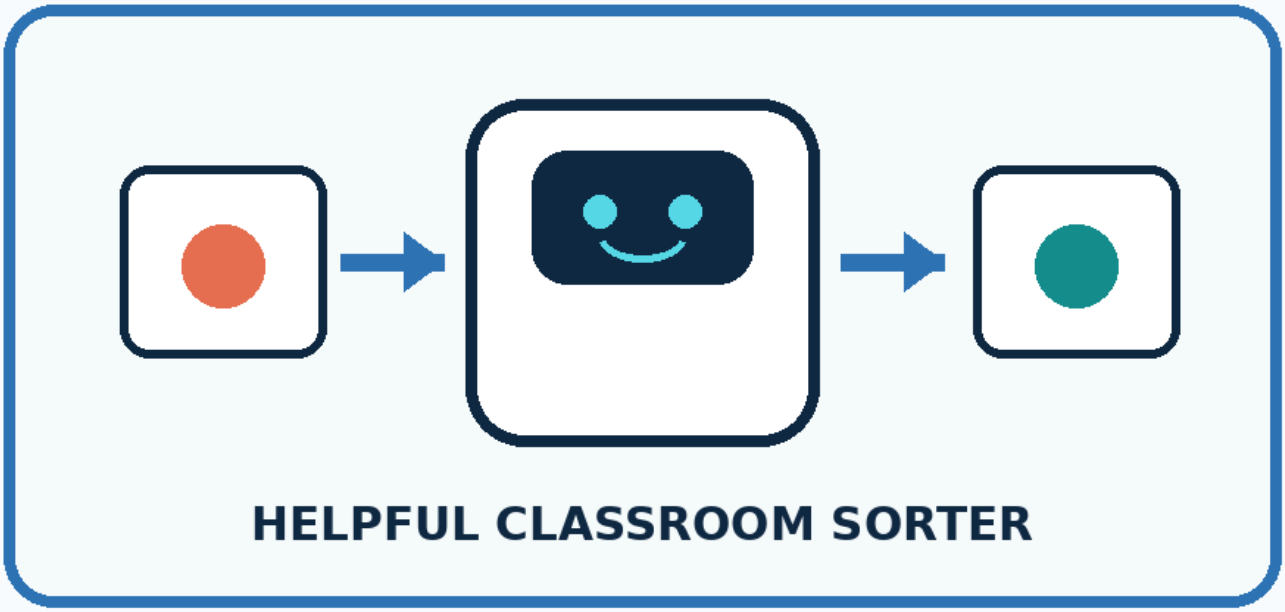
RESPONSIBLE DESIGN The **output** is not proof that the system understood the **input** as a person would.

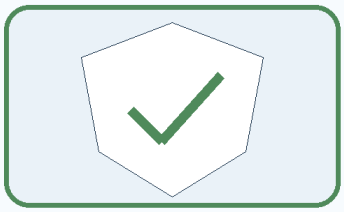
▼ **Earlier wording, alternative practice and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: trace, explain and name a loss

> Close your notes and give an explanation you can defend.

Your exit explanation

- Close your notes. Explain this goal in your own words: **Trace** information through a system without confusing information with a physical action

> **My deciding step, evidence and one remaining question**

Choose support and plan a return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use these words when they help you explain a particular arrow or missing detail.

Input, process, output and trace

Word	Meaning	My example
input	information entering a system	
process	the steps that transform information	
output	the result produced by the system	
trace	follow information through each stage	

Representation and information loss

Word	Meaning
Representation	A way of recording information so that it can be stored, communicated or processed
Information loss	Details that cannot be recovered from the retained representation

Repair a claim using a word

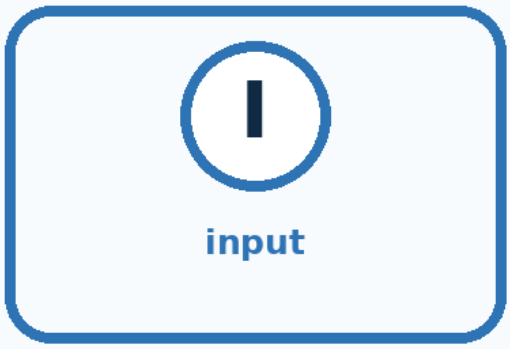
- CLAIM TO REPAIR The computer already knows the answer before it receives **input**.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ **Earlier wording, alternative practice and picture transcripts**

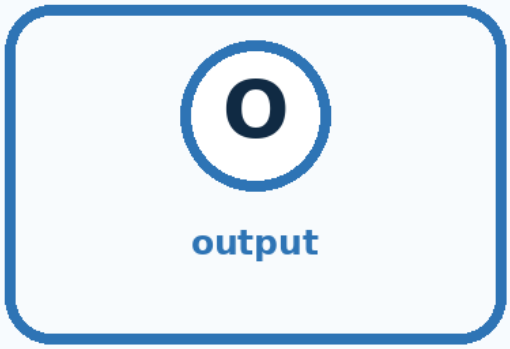
VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ **Choose a harder thinking challenge**

REFERENCE · SECTION 10

Choose a harder thinking challenge

> Choose one next step; you do not need to complete the entire ladder.

Pick a useful challenge

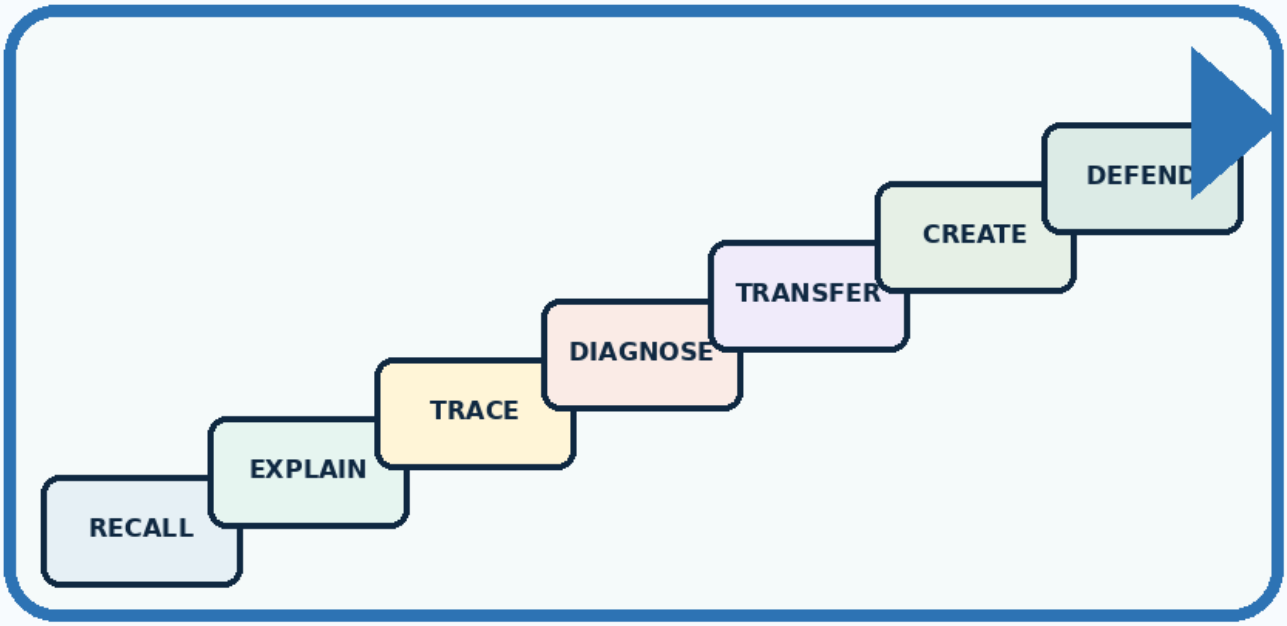
- RECALL Define **input**.
- EXPLAIN A system transforms information; it does not simply 'know.'
- **TRACE** Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: The computer already knows the answer before it receives **input**.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

<details>
<summary>Earlier wording, alternative practice and picture transcripts</summary>

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.
</details>



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check any information-path explanation**

REFERENCE · SECTION 12

Reference: check any information-path explanation

> Use after practice. The four-score case is this week's selected independent attempt.

Check your trace and its limits

- ☐ Name or draw the fresh case.
- ☐ **Trace input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record the support you used

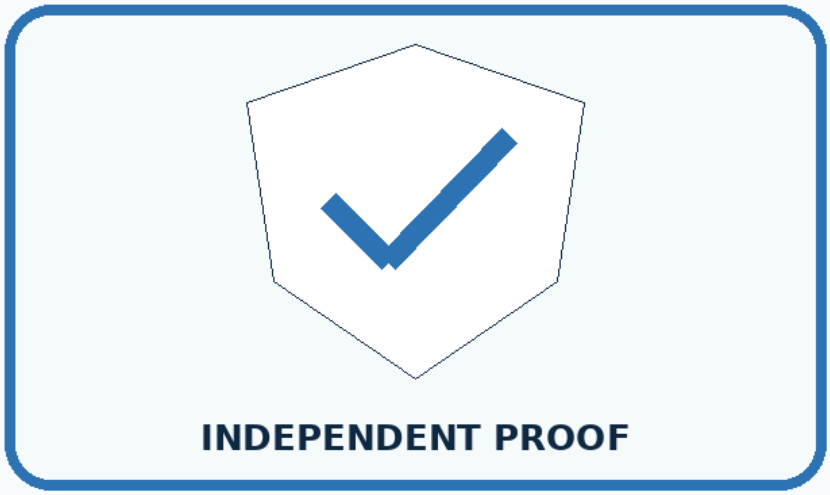
My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ **Earlier wording, alternative practice and picture transcripts**

MASTERY

Checklist for an information-path explanation

Use this checklist after practice. The separately named four-score case is this week's selected independent attempt.



EXPLANATION CHECK **Trace input**, **process** and **output**; explain a missing-**input** case and name a detail the stored representation cannot recover.

Independent **evidence**:

SELF-CHECK Start with what enters, name the actual transformation rather than saying the system thinks, and identify the result. If the **process** is unknown, state what **evidence** is missing instead of inventing a **mechanism**.

NEXT BRIDGE Carry today's **evidence** habit into: Sensors Are Messengers.

▼ **Extra practice: mood in, titles out**

REFERENCE · SECTION 13

Extra practice: mood in, titles out

> Try the playlist question before opening its worked explanation.

Read the case and question

A playlist helper receives a mood, matches song tags and displays three titles.

Name **input**, **process** and **output**. Is displaying titles the same as playing audio?

Try your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Input is mood; **process** is tag matching; **output** is titles. Audio playback is another step. Matching can use fixed rules; the description alone does not establish machine learning.

Evidence or stage	Value or result
Mood	Input from listener
Tag matching	Process chooses matches
Three titles	Displayed output
Audio playback	A separate possible step

▼ Earlier wording, alternative practice and picture transcripts

Week 2 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

From mood to song titles

A playlist helper receives a mood, matches song tags and displays three titles.

Name **input**, **process** and **output**. Is displaying titles the same as playing audio?

Input is mood; **process** is tag matching; **output** is titles. Audio playback is another step. Matching can use fixed rules; the description alone does not establish machine learning.

▼ Reference: the original lesson checklist

REFERENCE · SECTION 15

Reference: the original lesson checklist

> Use this if you want an additional checklist of this week's learning targets.

Targets and evidence

By the end, I can	How I will show it
Trace information through a system without confusing information with a physical action	Explain the deciding step, show a trace or test, and state one limit.
Explain which details a representation preserves or discards	Explain the deciding step, show a trace or test, and state one limit.
Locate the earliest stage at which an incorrect result could have arisen	Explain the deciding step, show a trace or test, and state one limit.

Keep your original prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternative practice and picture transcripts

Start with this question: A fictional classroom cupboard records coloured pencils, then reports just a total.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one paper or browser copy.

First supported question

- A fictional classroom cupboard records coloured pencils, then reports just a total.
- Which labelled step matches this detail? The saved totals cannot tell which box has more red pencils.
- 1. Add counts
- 2. Read colours
- 3. Ask a new question
- 4. Store total
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Ask a new question The saved totals cannot tell which box has more red pencils.

Adding preserves the number of pencils but discards how that number was divided between colours. Both distinct **input**s lead to the same stored **output**.

Working backwards from 6 therefore has several possible answers; it is not an instruction to divide equally.

Changed-record question

- Changed-case practice: Store red count alongside total. Which conclusion is supported by the supplied facts?
- 1. The records become (6,2) and (6,5).
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed record**

After your attempt, compare with the supported explanation: The records become (6,2) and (6,5). Blue count can now be recovered by total minus red, assuming these are the only colours.

Repair and save your work

Review AI or Not? (ai-1/1). Goal: Distinguish an observed result from an explanation of how it was produced. Short practice: Point to the learned stage. Does the changed display prove the weather forecast changed?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternative practice and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

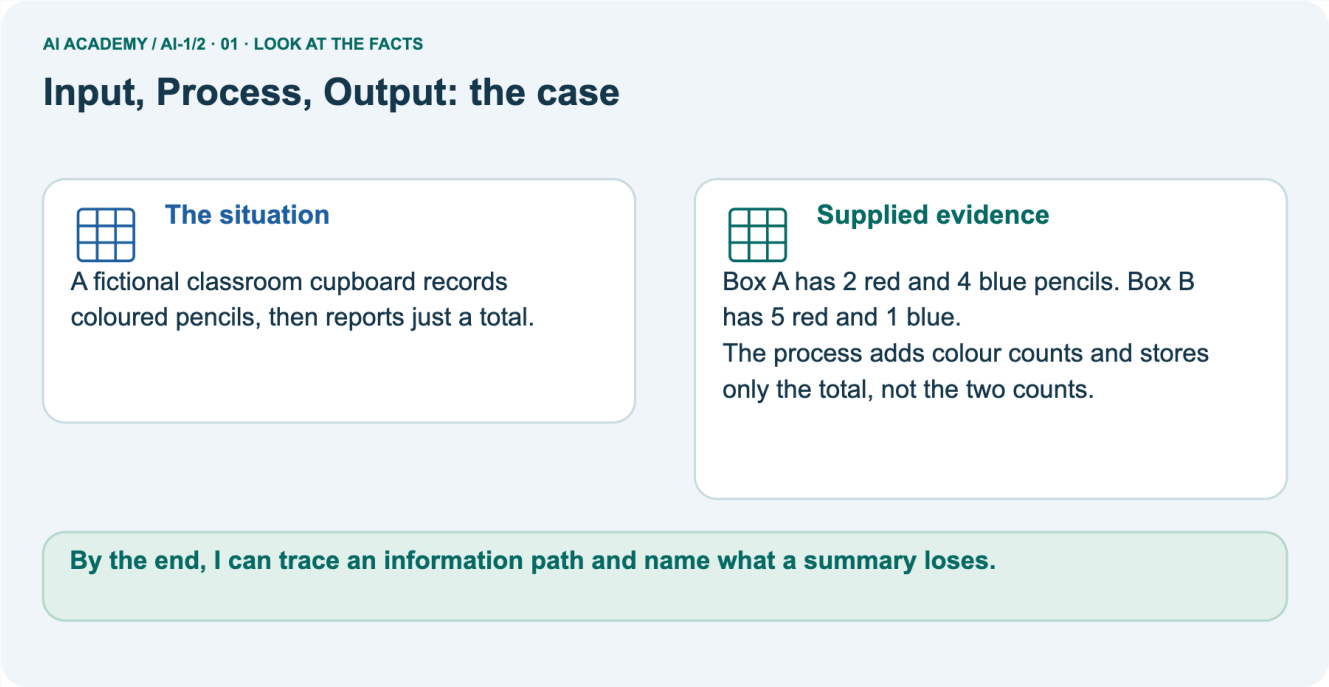
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the diagram that serves your current question; the main pictures already sit beside their explanations.

01. Input, Process, Output: the case



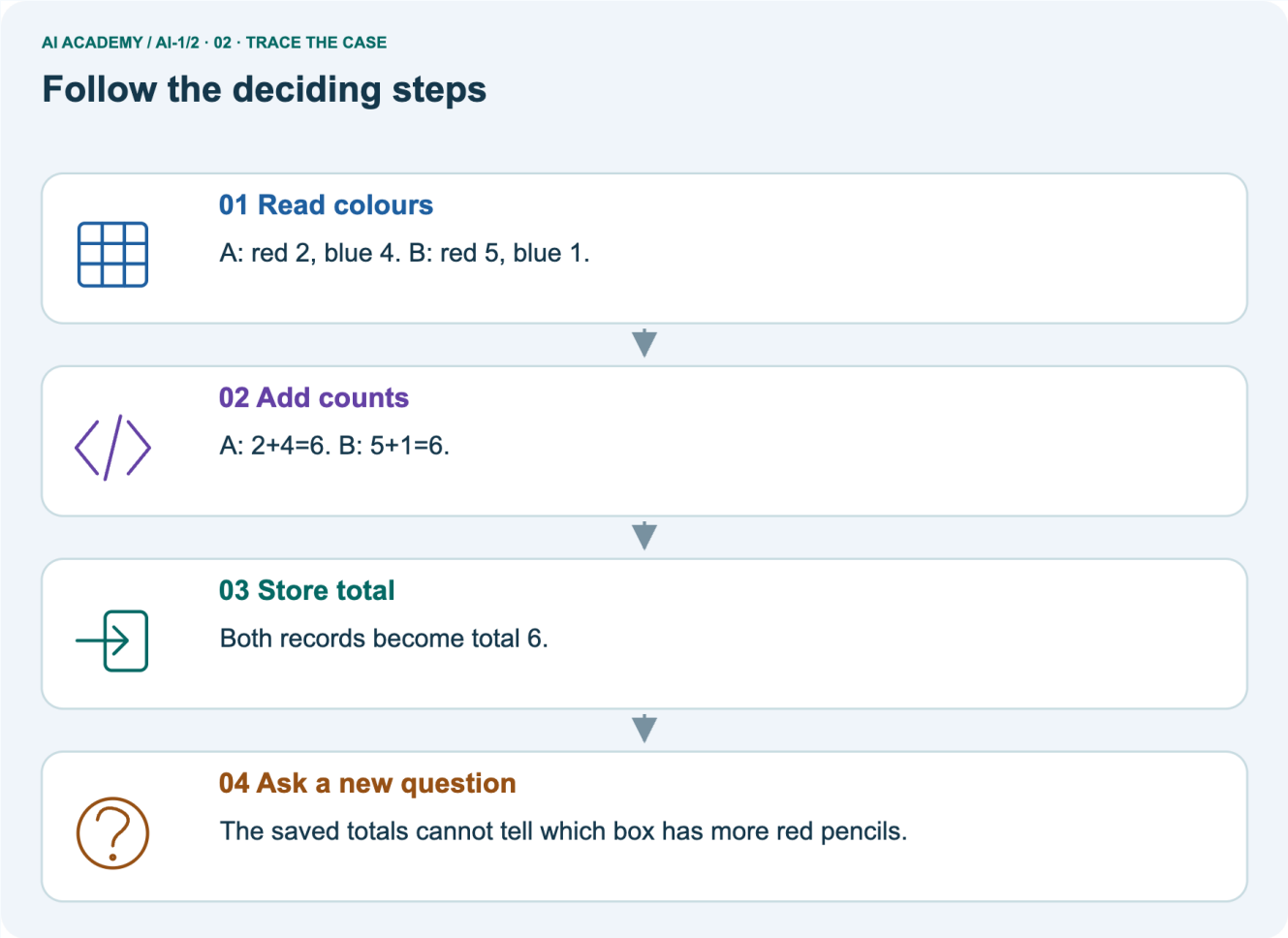
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional classroom cupboard records coloured pencils, then reports just a total. Supplied facts: Box A has 2 red and 4 blue pencils. Box B has 5 red and 1 blue.

The **process** adds colour counts and stores only the total, not the two counts.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

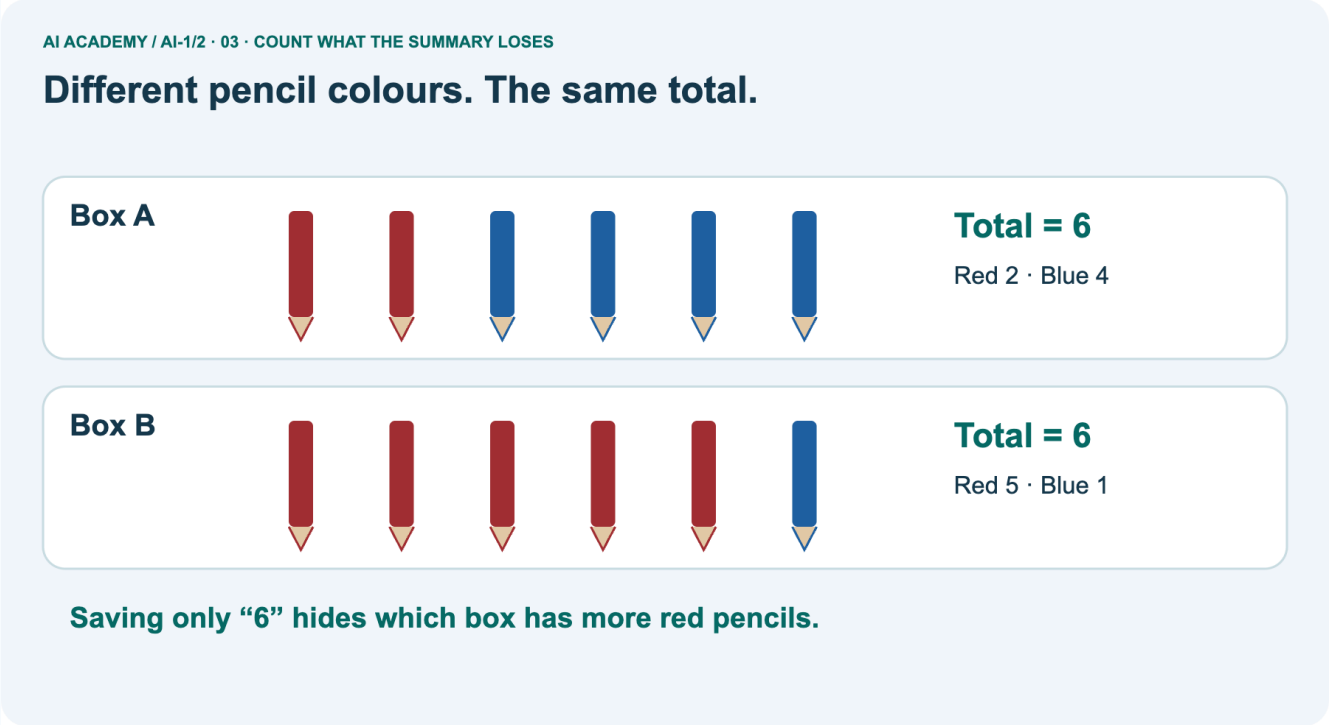


Follow the labelled picture with the matching explanation. Open the vector diagram

Read colours: A: red 2, blue 4. B: red 5, blue 1. → Add counts: A: 2+4=6. B: 5+1=6. → Store total: Both records become total 6. → Ask a new question: The saved totals cannot tell which box has more red pencils.

Point to each numbered stage and explain which recorded **input** it uses.

03. Different pencil colours. The same total.

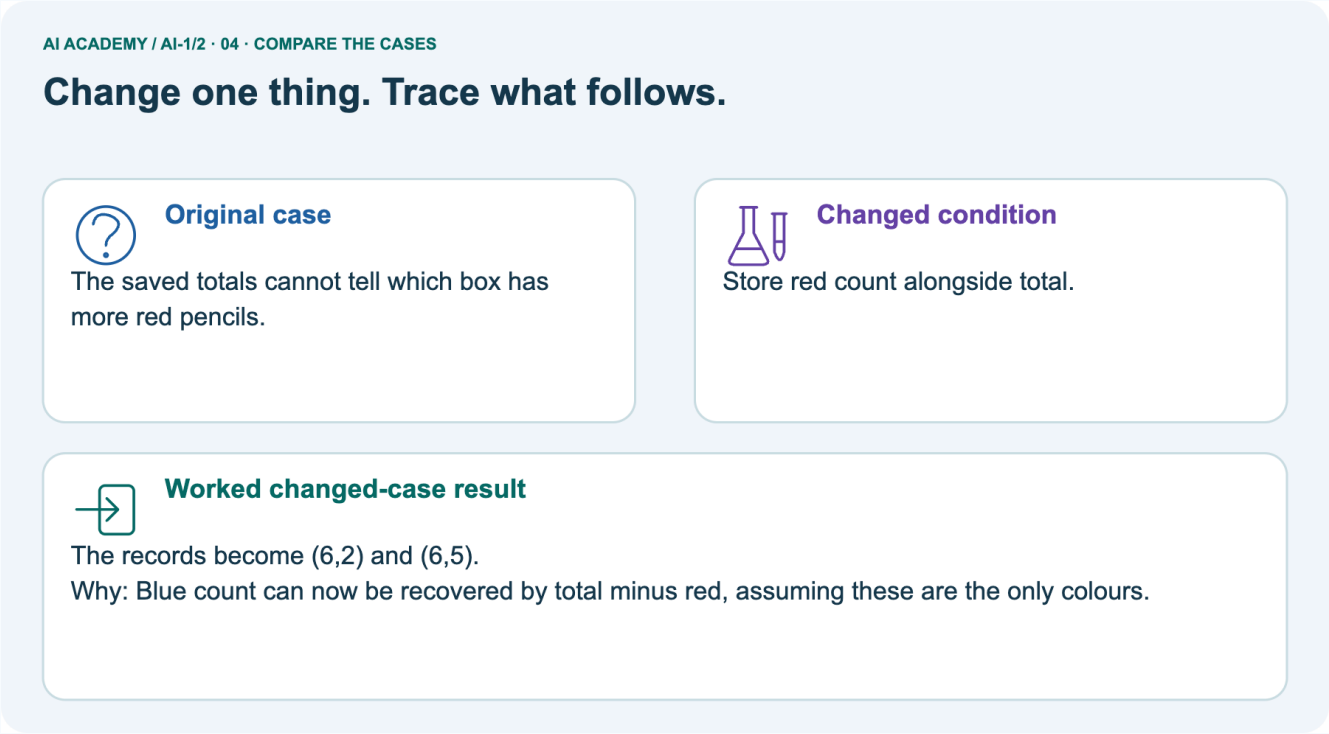


Follow the labelled picture with the matching explanation. Open the vector diagram

Box A: 2 red pencils and 4 blue, total 6. Box B: 5 red pencils and 1 blue, total 6. The equal totals do not preserve the colour counts.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



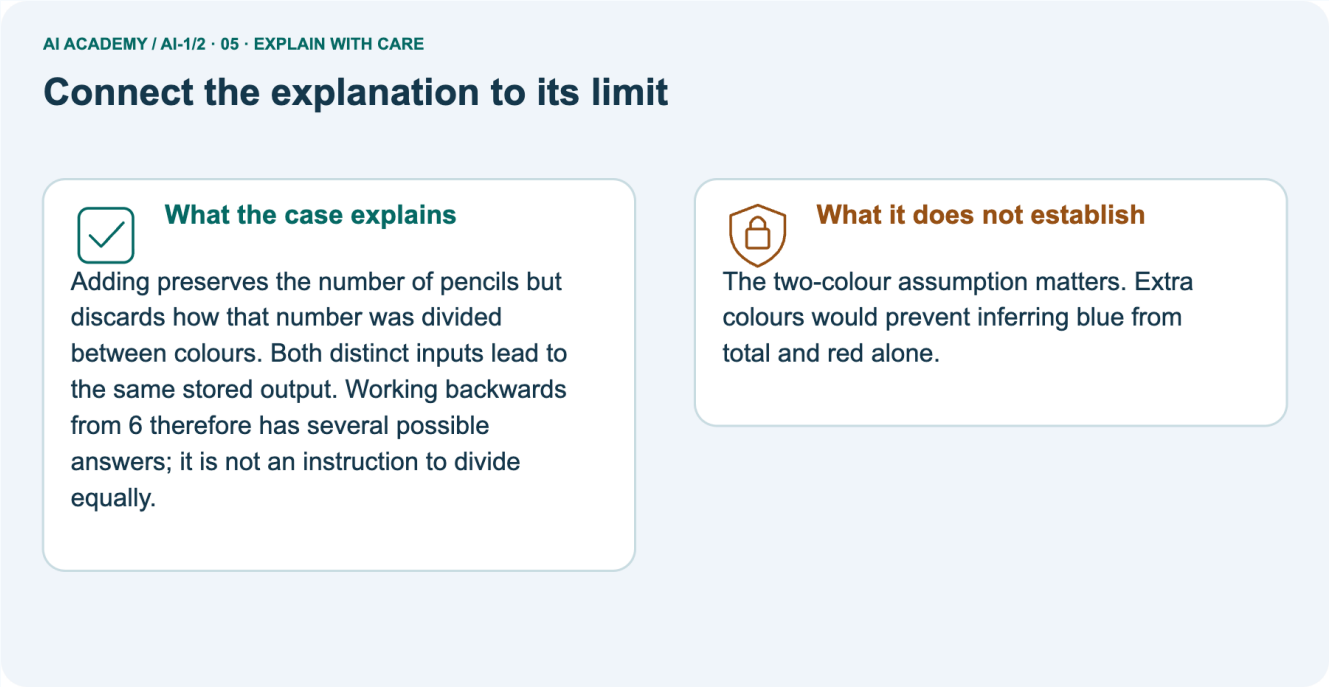
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The saved totals cannot tell which box has more red pencils. Changed condition: Store red count alongside total. Expected result: The records become (6,2) and (6,5).

Reason: Blue count can now be recovered by total minus red, assuming these are the only colours.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



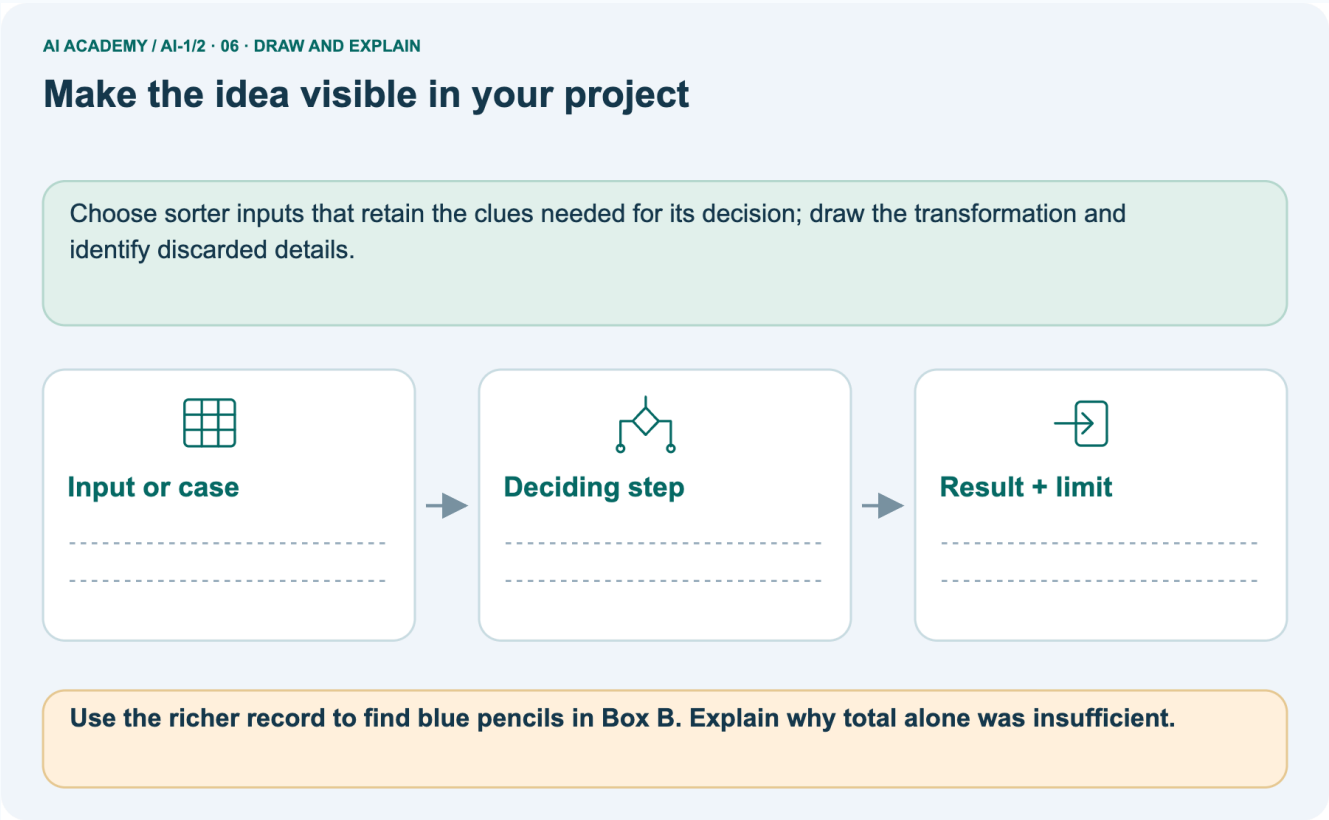
Follow the labelled picture with the matching explanation. Open the vector diagram

Adding preserves the number of pencils but discards how that number was divided between colours. Both distinct **input**s lead to the same stored **output**.

Working backwards from 6 therefore has several possible answers; it is not an instruction to divide equally. Limit: The two-colour assumption matters. Extra colours would prevent inferring blue from total and red alone.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose sorter **input**s that retain the clues needed for its decision; draw the transformation and identify discarded details. Draw your **input**, deciding step, result and limit.

Supported question: Use the richer record to find blue pencils in Box B. Explain why total alone was insufficient.

Draw your project **trace** and justify the deciding step using **evidence**.

▼ Earlier wording, alternative practice and picture transcripts

01. Input, Process, Output: the case

02. Follow the deciding steps

03. Different pencil colours. The same total.

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.